AI in Medicine HPML

Bryan Cardenas

High Performance Machine Learning

SURF

What are we doing @SURF?

01.

Histopathology

Detecting cancer and Precursor lesions

02.

Protein Design

Assist with protein design for vaccines

03.

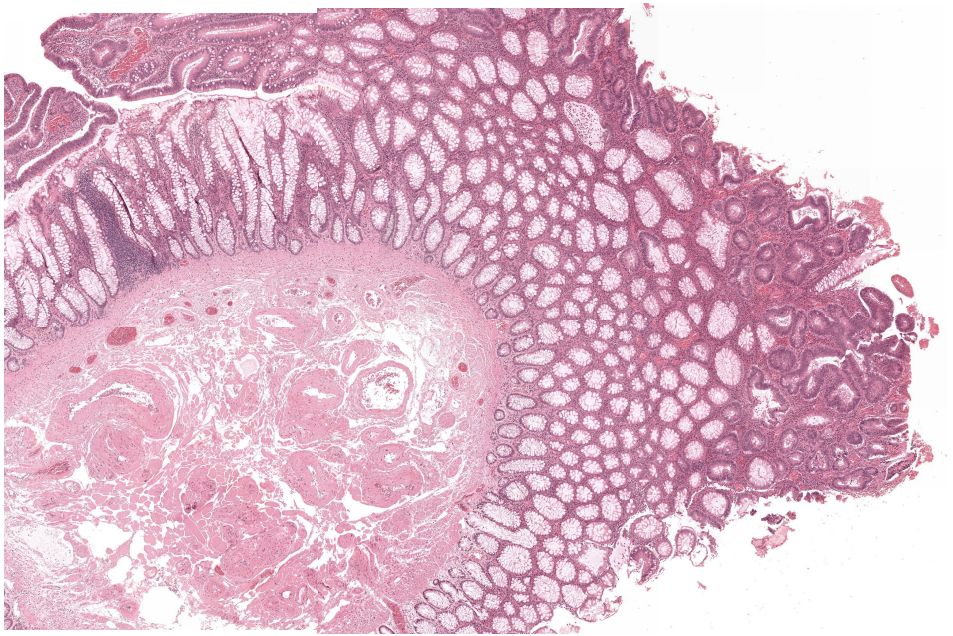
Hereditary

How to use Neural Networks in multi modal data environments across multiple labs in the EU.

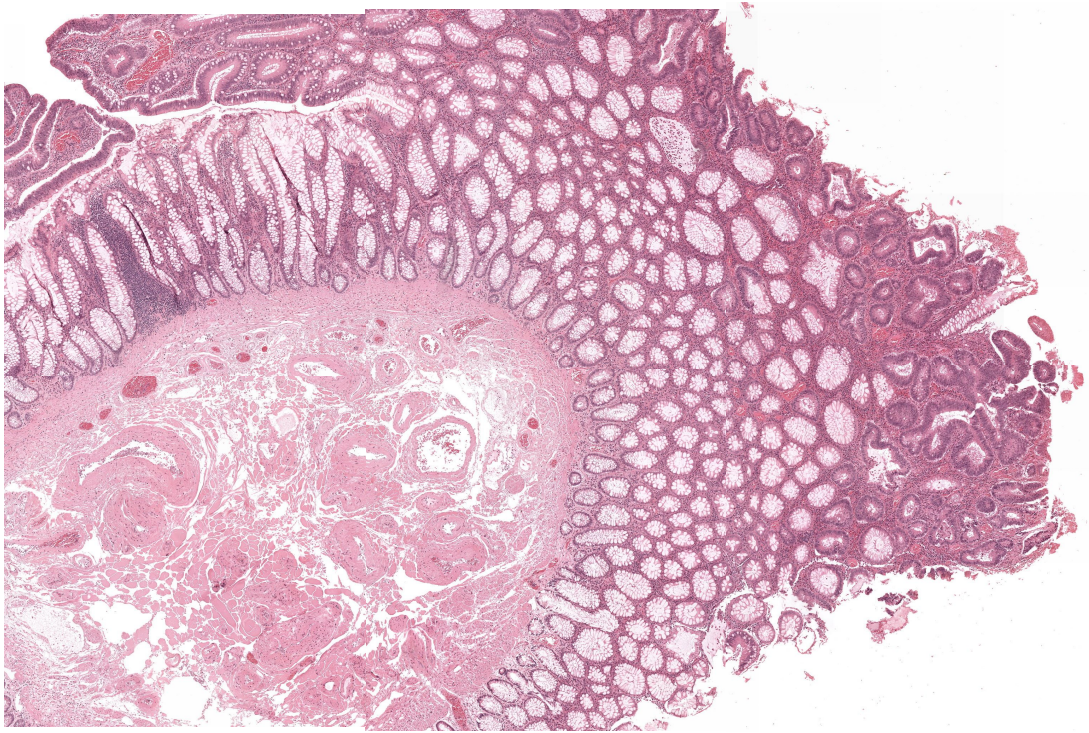
Histopathology

Histopathology Data Collection



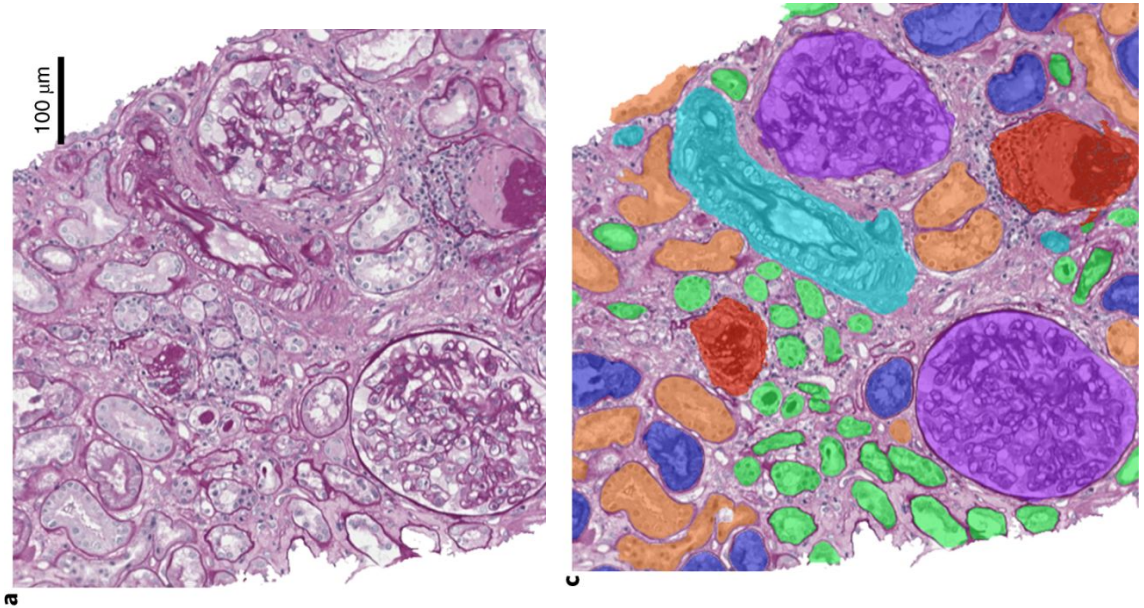


Histopathology Data: Images and Diagnostic Reports



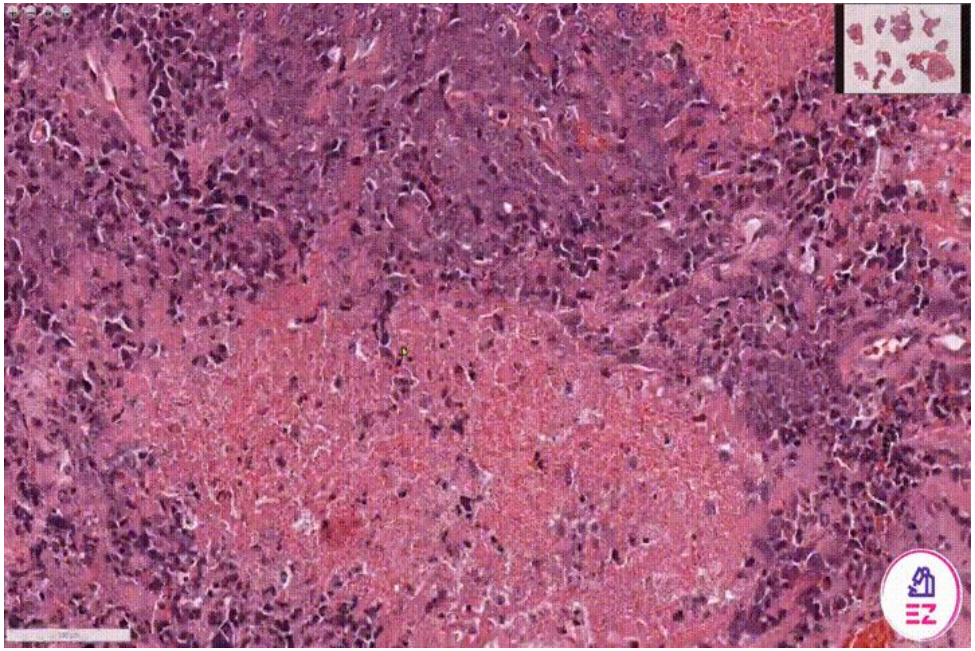
- “Colonbiopt 21-25 cm en 15 cm: in beide subrieken adenocarcinoma metastase, morfologisch en immuunhistochemisch geschikt voor het al bekend primaire ovariumcarcinoom van de patiënt.”
- 2 - adenoma tubulare con displasia bassa, focalmente, alto grado (moderato, focalmente grave)

Histopathology Data: Images and Diagnostic Reports



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Histopathology Data: Images and Diagnostic Reports



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https://huggingface.co/docs/transformers/model_doc/marian



Examode

01.

Data Management

02.

HPC Consultancy

03.

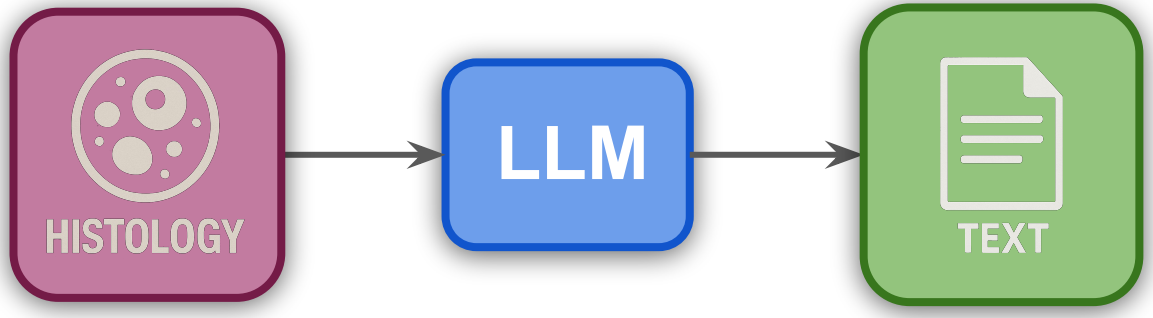
Classification, Segmentation, Multimodal Representation Learning

ExaMode

SURF

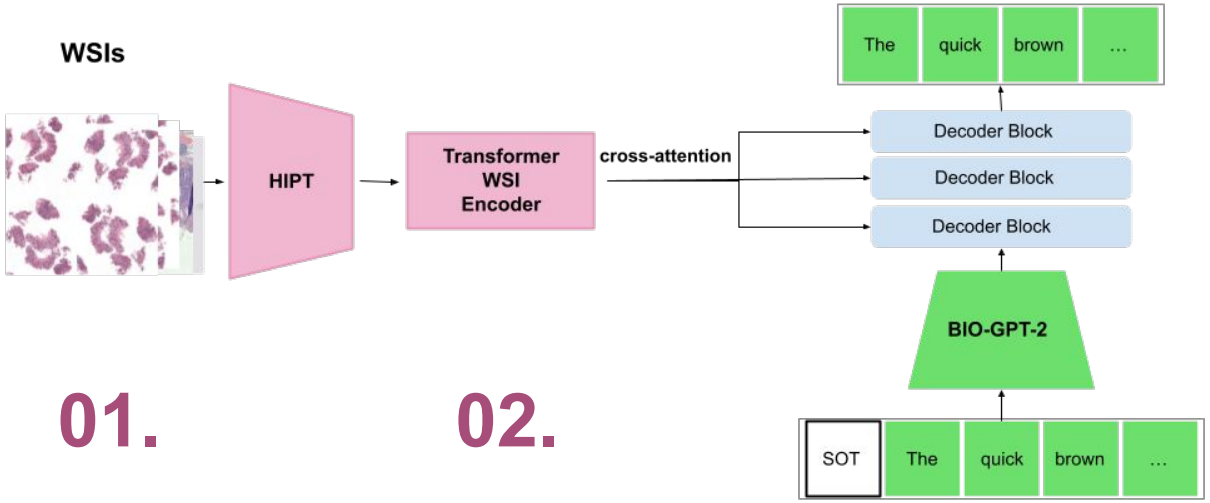


Histopathology
Conditioned Language Modelling





Histopathology Conditioned Language Modelling



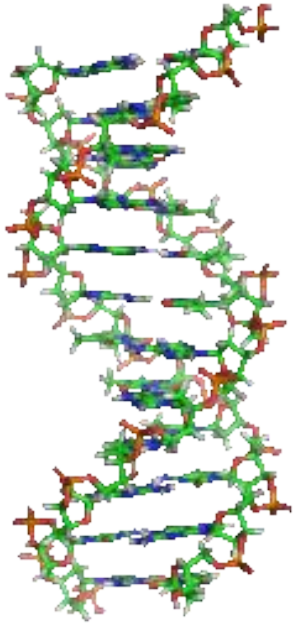
01. Data Management

02. Fitting generated text

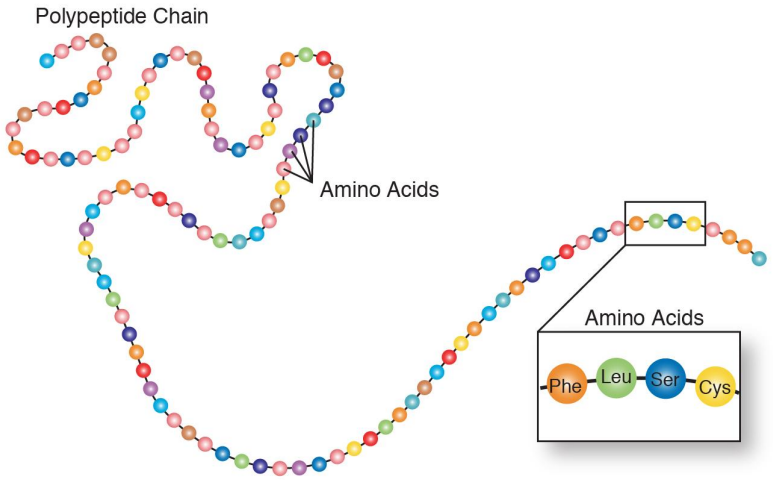
Protein Folding & Design

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High Performance Machine Learning Group



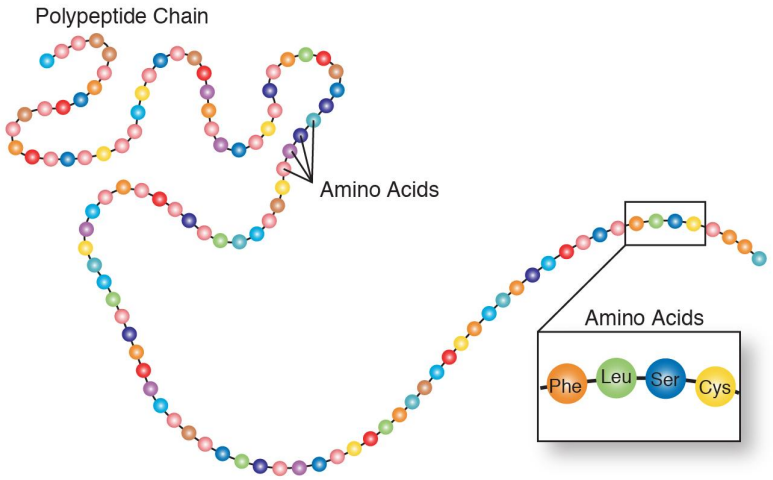
ACTG



Amino Acids

Ala: Alanine	Gln: Glutamine	Leu: Leucine	Ser: Serine
Arg: Arginine	Glu: Glutamic acid	Lys: Lysine	Thr: Threonine
Asn: Asparagine	Gly: Glycine	Met: Methionine	Trp: Tryptophane
Asp:Aspartic acid	His: Histidine	Phe: Phenylalanine	Tyr: Tyrosisne
Cys:Cysteine	Ile: Isoleucine	Pro: Proline	Val: Valine



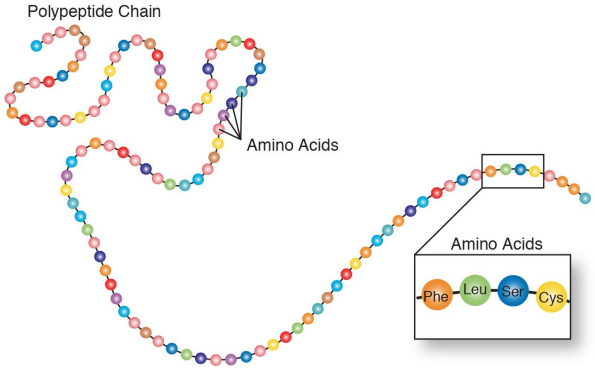


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AlphaFold2

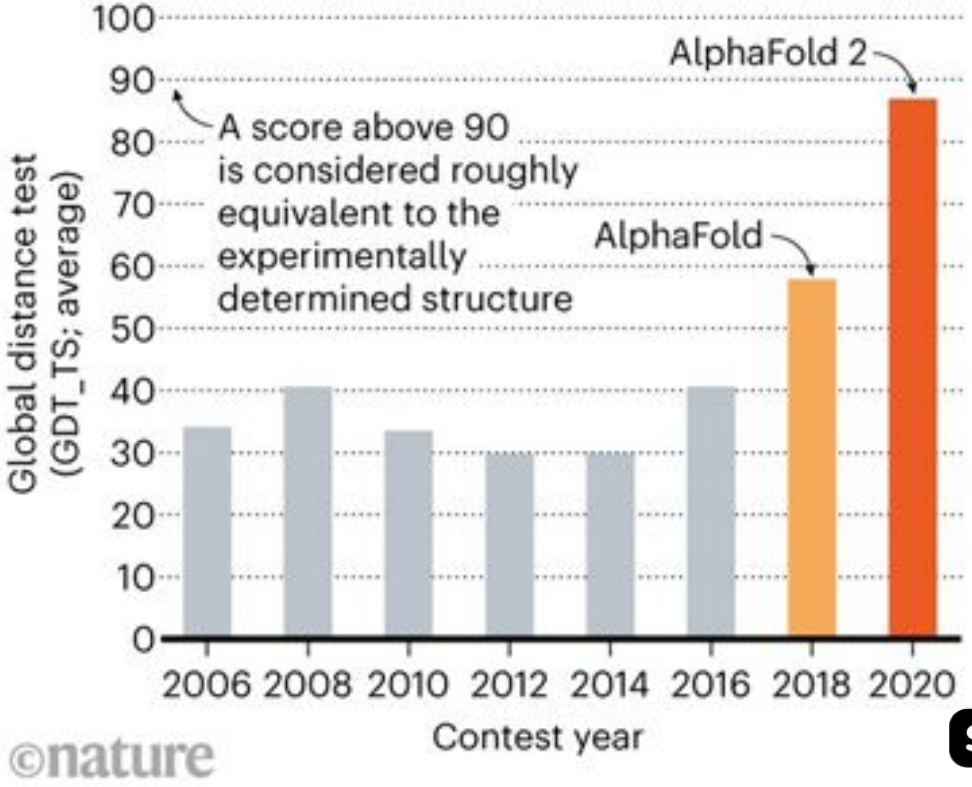




SURF

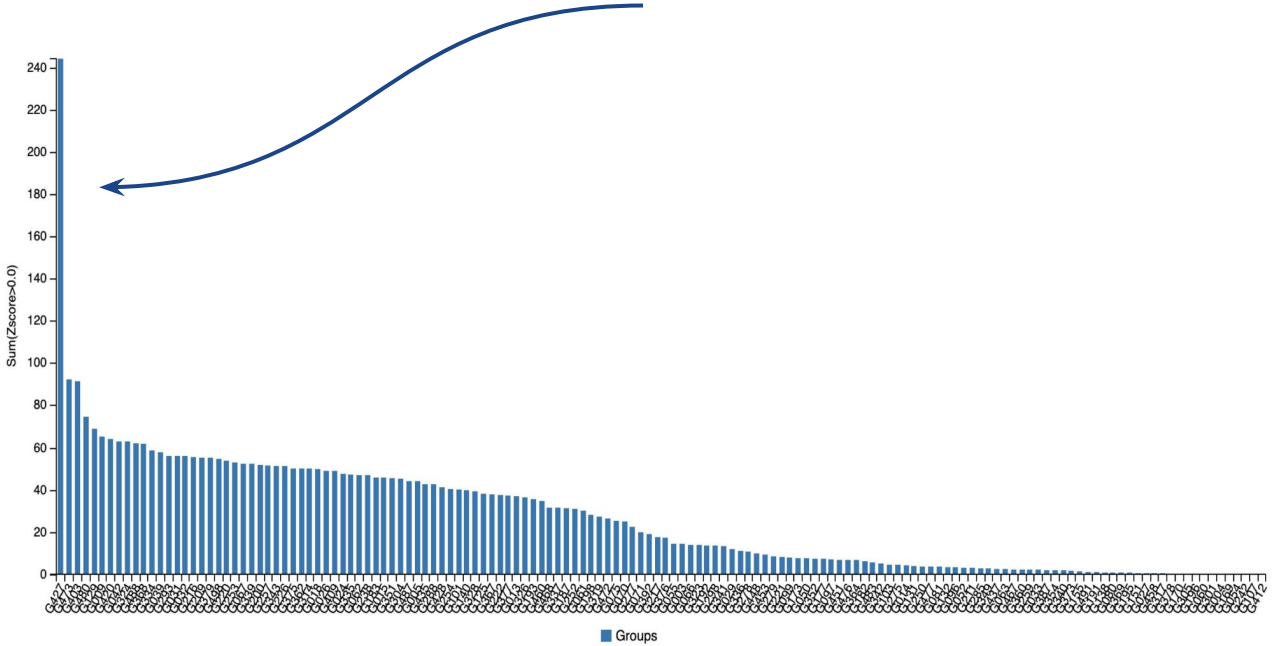


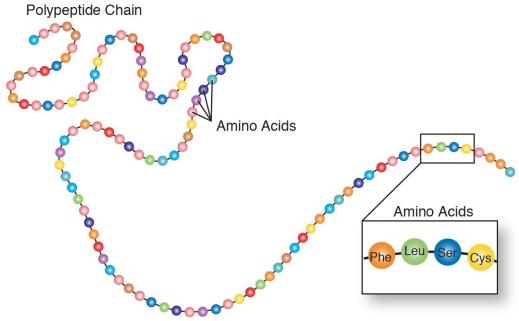
AlphaFold2


100
90
80
70
60
50
40
30
20
10
0
Global distance test (GDT_TS; average)
A score above 90 is considered roughly equivalent to the experimentally determined structure
AlphaFold 2
AlphaFold
2006 2008 2010 2012 2014 2016 2018 2020
Contest year
©nature

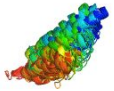


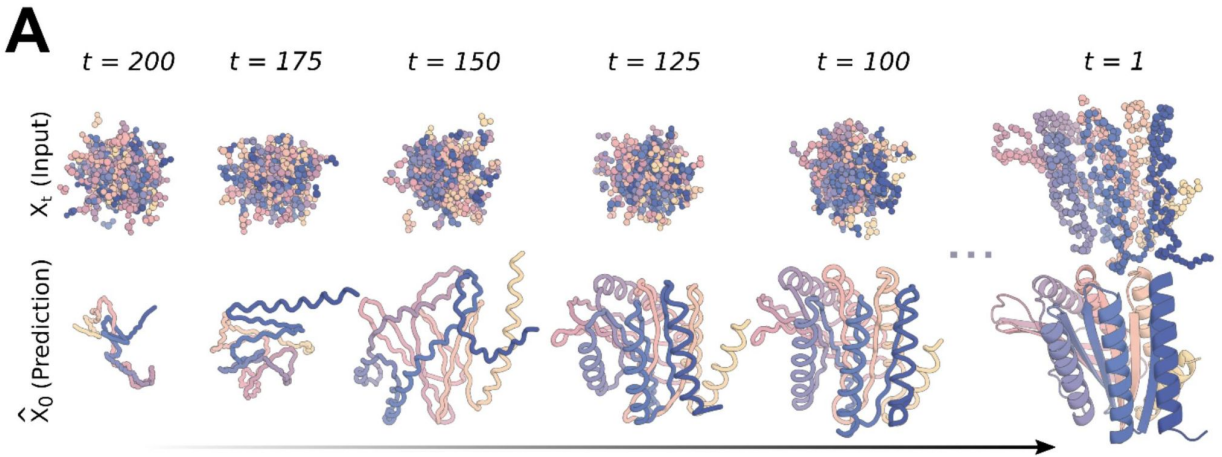
AlphaFold2


Sum(Zscore>0.0)
240
220
200
180
160
140
120
100
80
60
40
20
0
Groups


Polypeptide Chain
Amino Acids
Amino Acids
Phe
Leu
Ser
Cys

AlphaFold2

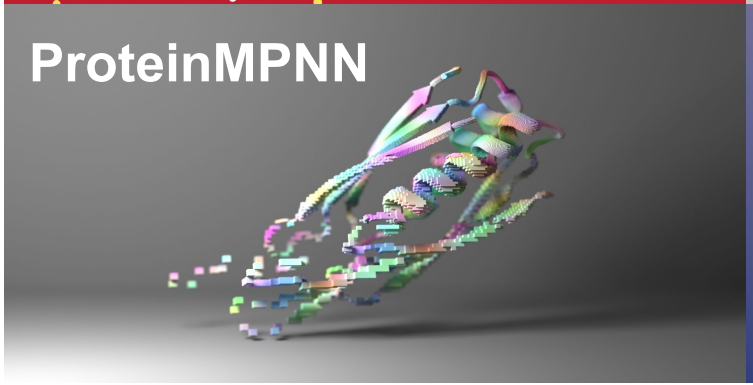


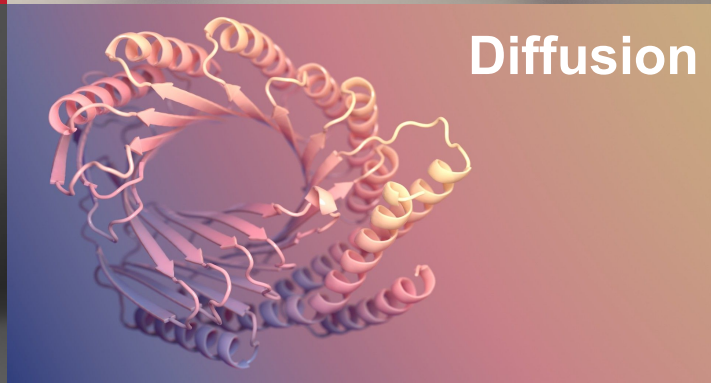

A
t = 200
t = 175
t = 150
t = 125
t = 100
t = 1
X_t (Input)
$\hat{X}_0$ (Prediction)

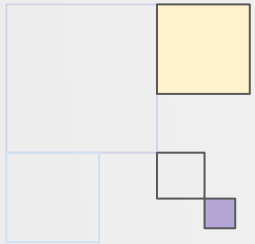
Suite of Protein Models



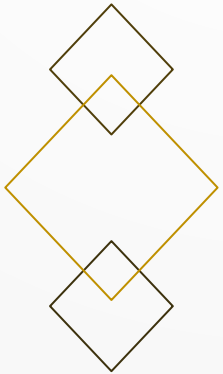








Experimental vaccinology@AMC

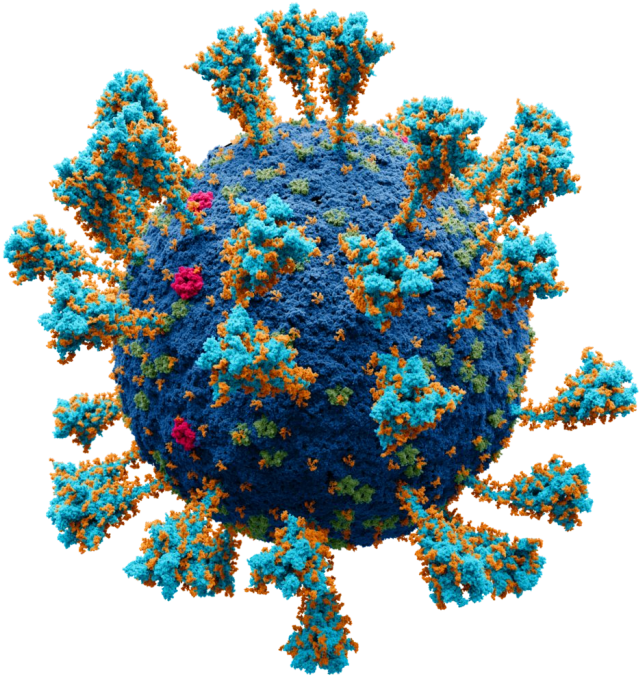


High Performance Machine Learning Group

SURF



SARS-COV-2

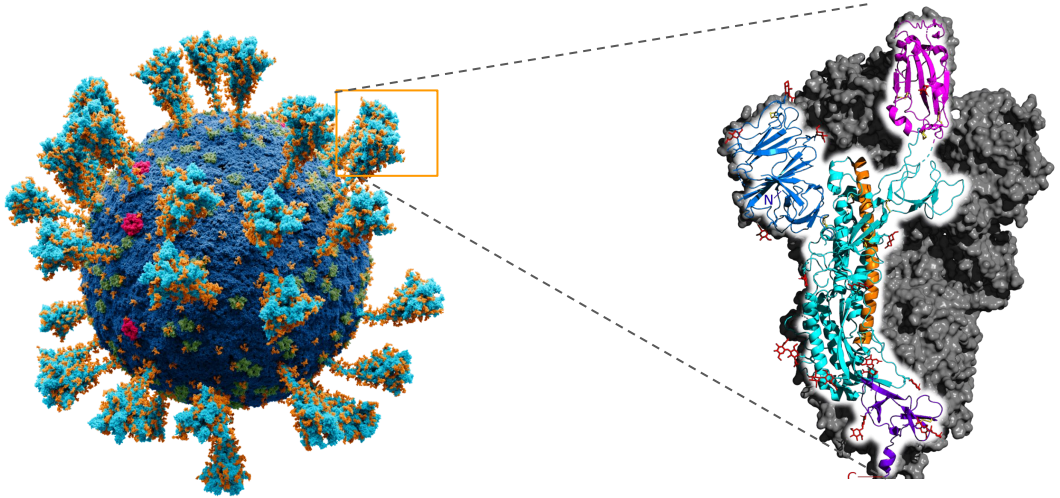


SURF



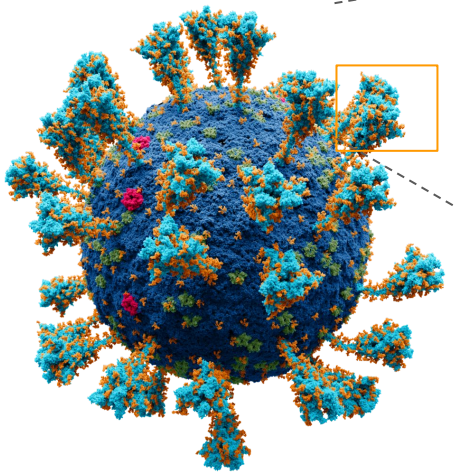
SARS-COV-2

Spike Protein

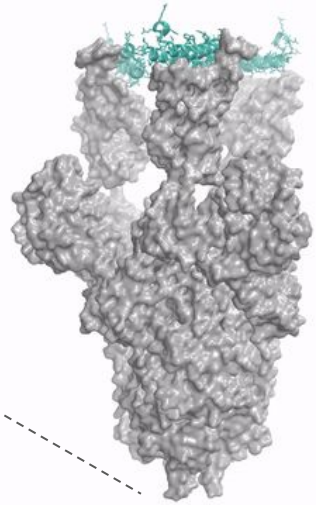




SARS-COV-2

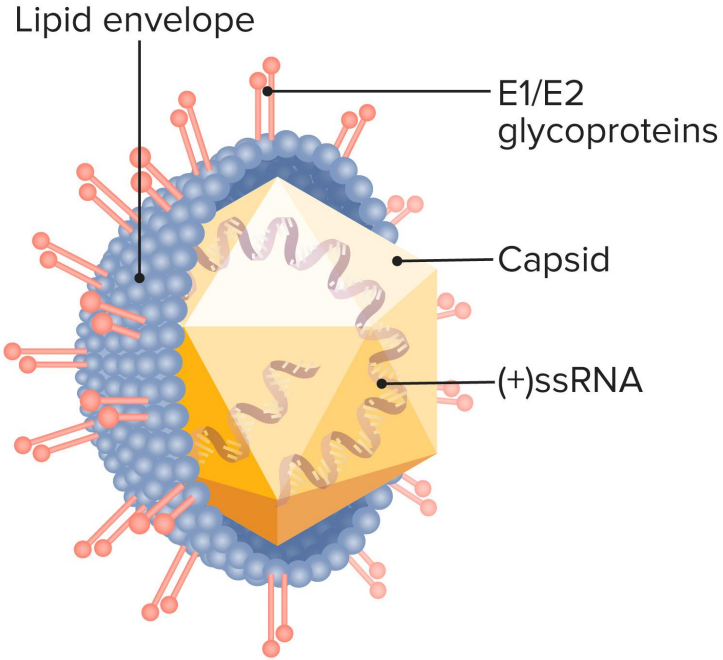


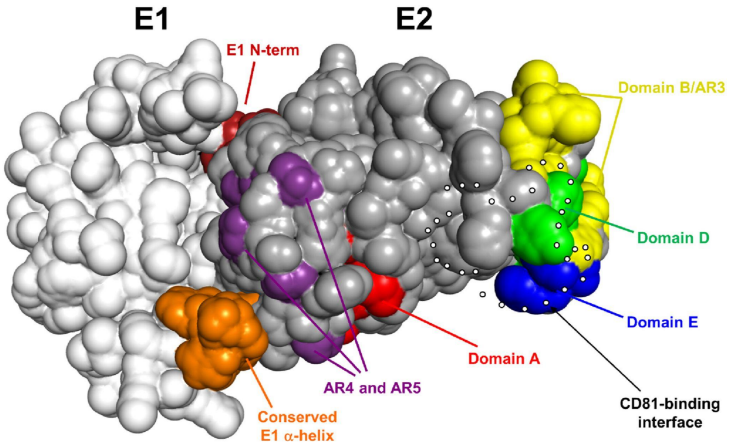
Spike Protein



Experimental vaccinology@AMC

Hepatitis C Virus





Recipe for Assistance

01.

Provide hardware & storage

02.

Run software

03.

Efficiency

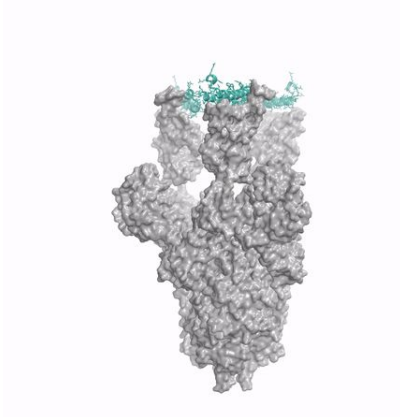
04.

Think of a new way to tackle the problem

Chaining Predictions

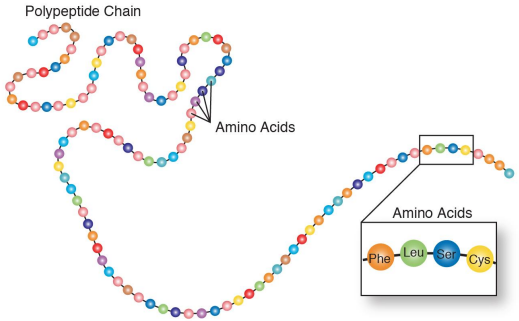
01. Generation

RFDiffusion



02. Predict Sequence

ProteinMPNN



03. Confirm Structure

ESMFold, ColabFold, AlphaFold, OmegaFold, RoseTTaFold



SURF

Hereditary



Hereditary

01.

Federated Learning

02.

Provide compute & HPC support

03.

Train Models on multiple modalities

04.

Security & Compliance monitoring System



Hereditary Data Types



SURF

Federated Learning

Improve diagnostic accuracy but maintain privacy!

Make data comply with privacy and ethics

Multiple centers benefit!





Gut-Brain Interplay

01. Neurological Diseases

02. Gut Microbiome Disorders

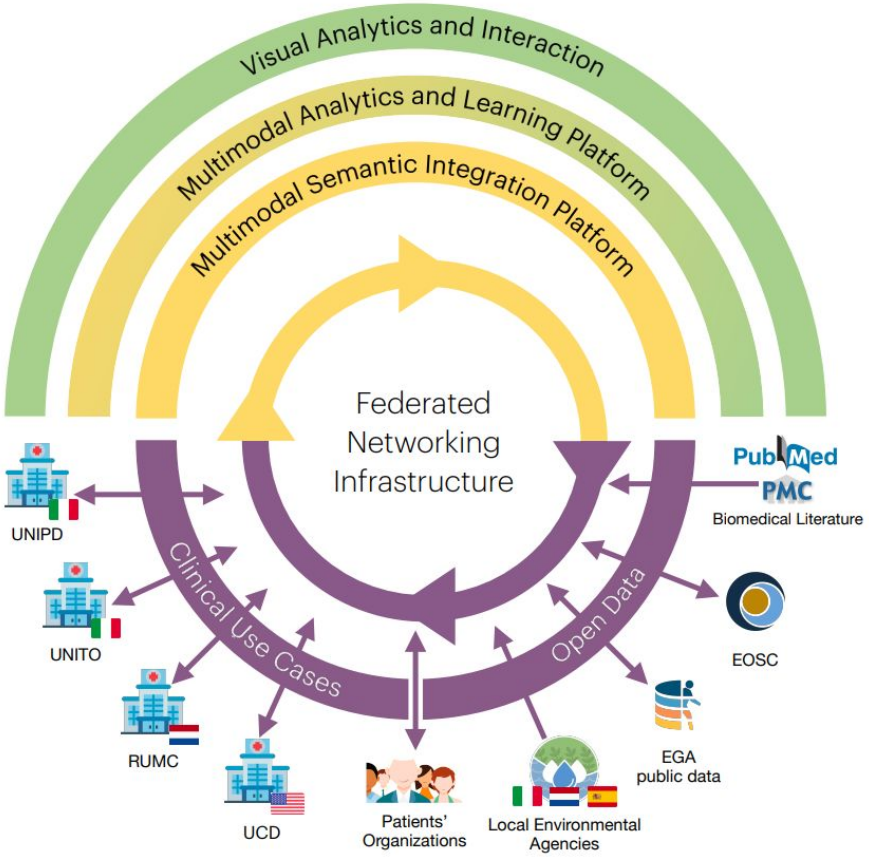
Multi-Modality
Multi-Linguality

Data Assembly
Privacy & Security

Funded by the European Union

HEREDITARY

The Call & Objectives



Secure & lawful federated access to linked multimodal data across Europe

Healthcare data normalization

Multimodal Ontologies

Federated Polystore for querying
Federated Learning Models

Analytics platform for aggregation and discovery from multiple sources

Patient Recommendations

Thanks!

Mail me for Questions:

bryan.cardenasguevara@surf.nl